

Immaculate Chukwuemeka
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PROFESSIONAL SUMMARY

Experienced Petroleum Engineer and Data Scientist with over 8 years of diverse experience in the energy and technology sectors. Proven track record of achieving significant production gains (e.g., 3,000 bopd boost) through strategic data analysis. Adept at leveraging advanced AI/ML capabilities, including predictive modeling, time-series analysis, deep learning, and data engineering, to drive data-driven optimization and enhance oil and gas production. Committed to advancing U.S. energy security and technological leadership and passionate about promoting community economic empowerment through digital literacy.

EDUCATION

Artificial Intelligence with Specialization in Data Analytics, MSc	(June 2026 - December 2027)
Indiana Wesleyan University Indiana, USA	
TechWise – Artificial Intelligence /Machine Learning & Web Development	(March 2024 - September 2025)
Talent Sprint Sponsored by Google Iowa, USA	
Computer Information Systems, AAS	(January 2023 - May 2025)
Des Moines Area Community College Iowa, USA	
Bachelor of Science in Petroleum Engineering – First Class Honors	(September 2007 - September 2011)
University of Ibadan Oyo State, Nigeria	

Certifications:

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|---|---------------------|
| • Integrated Oil & Gas Analytics using Big Data Virtual, Malaysia | (March – June 2025) |
| • Gas Monetization Workshop Accra, Ghana | (April 2022) |
| • Integrated Production Modelling Houston, USA | (March 2019) |
| • Well Production Enhancement London, United Kingdom | (December 2017) |

WORK EXPERIENCE

Evelyn K Davis Center for Working Families, Des Moines, IA	February 2023 – Present
Technology Specialist/ Digital Literacy Trainer:	
- Designed and delivered Microsoft Office training and one-on-one digital literacy support, successfully equipping over 400 low-income, underserved individuals with essential digital skills for clerical roles. - Managed the small business CRM system, ensuring data integrity and optimizing functionality for client management. - Executed strategic data analysis using CRM data, leading to a 70% improvement in client show-rate and boosting client workshop attendance by over 200% through targeted campaigns.	

- Supervised the design, development, and implementation of comprehensive computer training curricula and digital literacy programs, including daily lesson plans with differentiated instruction and tailored assessments, ensuring alignment with key objectives for diverse student needs.

Sepal AI, San Francisco, CA

January – March 2026

Petroleum Engineer (Contractor) | Remote:

- Develop expert-level Petroleum Engineering datasets and evaluation tasks used to train and benchmark advanced large language models (LLMs) and AI systems.
- Design complex engineering scenarios covering production optimization, reservoir management, artificial lift design, well intervention planning, and hydrocarbon recovery strategies.
- Generate high-quality ground-truth responses and technical reasoning workflows that improve AI model accuracy in specialized engineering domains.
- Evaluate AI-generated outputs for technical correctness, completeness, and adherence to industry engineering standards.
- Leverage over eight years of petroleum engineering experience to bridge domain expertise and artificial intelligence applications.
- Support the development of AI systems capable of performing sophisticated engineering analysis and decision-support functions.

Valon Consulting Group, Houston, Texas

May – August 2025

Full Stack Intern:

- Developed and optimized 3 websites and applications, leading to a 15% increase in page load speed and significantly enhancing user experience.
- Collaborated with cross-functional teams to align web content with branding and marketing goals, contributing to a 10% increase in lead generation.

First Hydrocarbon Nigeria, Lagos, Nigeria

November 2016 – August 2022

Associate Production Engineer:

- Led the revamp and optimization of a gas lift injection system, contributing to a direct production uplift of circa 3,000 bopd.
- Contributed to well surveillance processes across multiple assets, identifying and executing production improvement opportunities on wells, facilities, and operations, which contributed to an additional circa 1,500 bopd in production.
- Contributed to achieving a 43% cumulative annual production growth rate increase without drilling new wells, accomplished through rigorous shut-in well reviews and rapid execution of identified remediation options.

- Designed, implemented, and optimized artificial lift systems for diverse horizontal and vertical wells with varying depths and fluid types, ensuring sustained production.
- Routinely monitored daily well performance, identified changing well conditions, and provided actionable recommendations to optimize production, supported by dynamic well models and historical data analysis to identify oil gain opportunities.
- Orchestrated well and reservoir surveillance programs, managing critical datasets (e.g., BHP, MDT, PVT, RST, PLT) and prioritizing candidates for workovers, stimulation, artificial lift, and compression to maximize asset performance.

Kogzytech Associates Limited, Lagos

October 2015 – September 2016

Technical Assistant to Principal Safety Consultant:

- Provided direct support to the Principal Safety Consultant in preparing detailed safety reports, risk assessments, and presentations for management and regulatory bodies, ensuring clear communication of safety performance and initiatives.
- Authored and implemented 15+ Standard Operating Procedures (SOPs) for routine field activities, enhancing the safe and effective operation of the process plant and wellheads.
- Developed and managed a comprehensive database for the safe handling of over 50 distinct production chemicals in the field, significantly improving product stewardship and workplace safety for field personnel.

Afren Nigeria PLC, Lagos, Nigeria

June 2013 – September 2015

Production Trainee Engineer:

- Conducted daily production surveillance, identifying trends and anomalies to support continuous operational efficiency.
- Performed feasibility studies on production data, applying various choke correlations to accurately predict well future response to choke changes, optimizing production outcomes.
- Contributed to process and production optimization initiatives, identifying opportunities to enhance output and operational efficiency.
- Managed field-produced water and gas lift optimization efforts, ensuring efficient resource utilization and production sustainability.
- Executed production accounting functions, including accurate back allocation and deferment tracking, to support robust reporting and planning.

PAPER AND PRESENTATIONS

- **Paper, SPE Nigeria Annual International Conference and Exhibition** (August 2020)
Title: *Unlocking Potential in Mature Brown Field with Complex Configuration in the Niger Delta- The OML26 Success Story.*

- **Paper, SPE Nigeria Annual International Conference and Exhibition** (August 2019)
Title: *Production Optimization Strategies for Unlocking Resource Potential in A Heavy-Oil Brown Field*
- *The OML26 Success Story*.

PROJECTS

Oil Production Prediction using AI/ML

<https://github.com/ImmaculateOkoruwa/Oil-Production-Prediction-AI-ML/>

- Applied machine learning algorithms to predict oil production based on historical data.

Bank Customer Churn Prediction| Machine Learning

<https://github.com/ImmaculateOkoruwa/TechWise-Machine-Learning-Project/>

- Built a predictive model using scikit-learn to identify high-risk customer churn.
- Utilized feature engineering and classification algorithms to improve business retention strategies.

Tetris Game | Python – Pygame, Object-Oriented Programming

https://github.com/Alfie-1516/Cohort_3_Tetris/tree/master

- Collaborated with a team to develop a Tetris game in Pygame, implementing dynamic gameplay, score tracking, level progression, and custom graphics.

SELECTED AWARDS

- **New Century Workforce Pathway Scholar representing the state of Iowa** **2025**
Phi Theta Kappa Honor Society and The Coca-Cola Foundation
Selected as Iowa State Representative over 2,100 applicants
- **Outstanding Commitment Award** **2024**
TechWise Talent Sprint Sponsored by Google | Iowa, USA
- **President Honor's List (Awarded 6 times)** **2023-2025**
Des Moines Area Community College (DMACC)

SKILLS

• Data Science & AI/ML	• Machine Learning (ML)
• Production Optimization	• Risk Assessment, Mitigation, and Controls
• Data Analysis & Insights	• Strategic Planning
• Operational Management	• Operational Excellence and Innovation
• Leadership and Communication	• Cross-functional Collaboration
• Project Management	• Problem-Solving
• Risk & Compliance	• Digital Literacy Programs